

Official Statistics

Mid-Semester Revision Notes: the System of National Accounts

B.Stat. 3rd year, Semester V — Lectures 1–12, keyed to the thrust areas and past papers

These notes cut the 62-page lecture compilation down to what the thrust-area list names, plus two items the list omits but the 2025–26 paper needs: taxes and subsidies, and residence, which GNI depends on. The history of economic thought (Lecture 1–2: mercantilists, physiocrats, Smith, Say) is dropped. It is on neither the list nor any paper.

Each section opens with a grey line giving its thrust areas (T1–T26, numbered as on the list), its lectures, and the past-paper parts it answers. **Orange boxes matter most.** The 2025–26 Q2 took two worked examples from the slides and replaced the numbers with symbols. Every slide example is therefore given here in both forms. Model answers to the paper are in §17.

T	Thrust area	§	Asked in
1	GVA, GVO, GDP, HFCE, IC, GCF, GNI — definitions and relations	1, 8, 11, 12	25–26 Q1(i), (ii), (iv); Q2(i)
2, 3	Two-sector circular flow (real and money flows); $Y = C + I$	1, 14	—
4, 5	Economic flows; exchange, transfer, transactions	2	—
6	Current and capital transactions	2	25–26 Q1(viii)
7, 8	General production boundary: inclusion/exclusion; IPP	3	—
9, 13, 14	Economically significant prices; market, non-market output, final use	5	—
10	Depreciation, depletion, savings	9, 12	25–26 Q1(vi)
11	Purchasers', producers' and basic prices	6	25–26 Q1(iii)
12	External trade exclusions	7	—
15, 17	Operating surplus (listed twice)	8	25–26 Q2(i)
16	Earned income	11	25–26 Q2(ii)
18, 19	Transfers; disposable income	12	—
20, 22	Financial and non-financial assets; inventories	13	25–26 Q1(v)
21	GDCF, GFCF, HFCE, GFCE	12, 13	25–26 Q1(iv)
23	Production, income and expenditure identities	14	—
24	Current account, accumulation account, balance sheet	15	—
25, 26	Monetary/non-monetary transactions; SNA and MTA	2, 16	—
—	<i>Not on the list:</i> taxes and subsidies	10	25–26 Q1(vii)
—	<i>Not on the list:</i> economic territory, residence, exports and imports	4	25–26 Q1(i)

How the two past papers line up with this syllabus

2025–26 midsem (30 marks) fits the thrust areas completely. Q1 is 25 marks: any 5 of 8 pairs, each marked 2.5 + 2.5, so define both terms, then state how they are related. Q2 is 5 marks, and both options are slide examples with symbols in place of numbers.

2024–25 midsem asked about SDGs, the health classifications ICD, ICHI, ICF and ESPAR, and DPSIR and FDES environment statistics. **None of this is on this year's list or in the notes.** That paper followed a different syllabus order: its endsem covered FDES, the World Health Survey and similar topics. Nothing in these notes addresses it.

What to expect this year

Q1-type pairs the list names but 2025–26 did not ask: GDCF and GFCF; market and non-market output; earned and disposable income; GNI and GNDI; financial and non-financial assets; exchange and transfer; monetary and non-monetary transactions; current account and accumulation account; IC and final consumption; rent and rental; economic flows and transactions.

Q2-type slide examples not yet used: the household's disposable income and saving (§12); pen and school prices (§6); the farmer's FCE and saving (§12); the NPI's borrowing (§13); the 18,000 economy (§1); the wallet and household MTA, and the MTA-to-accounts chain (§16).

1 Production, value added and the circular flow

Thrust areas T1, T2, T3 · Lectures 1–2 · PYQ: 2025–26 Q1(i), (ii)

Production is measured without double counting. Marshall's example is a carpet: once the carpet is counted at full value, the yarn and the labour in it are already counted. A producer is therefore credited only with the value it *adds*.

- IC** **Intermediate consumption:** goods and services a producer buys and *uses up during* the production process of the period, such as raw materials, fuel, power, maintenance and hired services.
- GVO** **Gross value of output:** the value of all goods and services the unit produces in the period.
- GVA** **Gross value added** = $\text{GVO} - \text{IC}$: the unit's own contribution to production.
- GDP** **Gross domestic product:** the sum of GVA over all resident producers, plus net product taxes. This is the value of production within the economic territory.
- HFCE** **Household final consumption expenditure:** what resident households spend on goods and services for final consumption.
- GCF** **Gross capital formation**, or investment: acquisitions less disposals of produced assets, namely fixed assets (GFCF), change in inventories and valuables.
- GNI** **Gross national income** = $\text{GDP} + \text{net primary (earned) income from the rest of the world}$. This is the income of residents.

Relations to reproduce

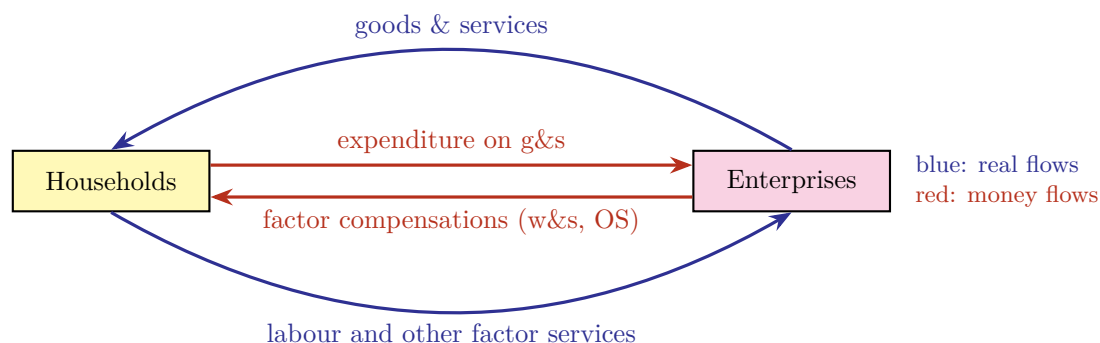
$$\text{GVA} = \text{GVO} - \text{IC}, \quad \text{GVA}_{bp} = \text{GVO}_{bp} - \text{IC}$$

$$\text{GDP} = \sum \text{GVA}_{bp} + (t - s)_{\text{products}} + (t - s)_{\text{imports}}$$

$$\text{GVO} = \text{HFCE} + \text{IC} + \text{GCF} \implies \text{GVA} = \text{HFCE} + \text{GCF}, \text{ i.e. } Y = C + I \quad (\text{closed, no govt.})$$

$$\text{GNI} = \text{GDP} + \text{net primary income from RoW}, \quad \text{Net} = \text{Gross} - \text{D\&D}$$

The two-sector circular flow (T2)



- Households own *all* factors of production (land, labour, capital, entrepreneurship). They sell factor services and are paid factor compensations: wages & salaries and operating surplus. Those payments are the economy's income.
- Real flows (goods and services, factor services) run opposite to money flows (expenditure, compensations).
- The model rests on Say and Walras: there is no lag between production and income, and no income goes unspent. So income equals expenditure (Say's law, "supply creates its own demand"). Keynes (1936) rejected this: spending depends on income, effective demand drives output, and aggregate expenditure is $C + I$.
- Hence **production** $\equiv$ **income** $\equiv$ **expenditure**, which gives the three ways of measuring GDP. The *production* approach sums GVA over all producers. The *income* approach sums the incomes generated. The *expenditure* approach sums *final* expenditure on the goods and services produced.

$$Y = C + I \text{ (T3)}$$

The simple circle leaves out enterprise-to-enterprise flows: inputs (IC) and capital goods (investment). Once they are added, the inputs cancel in aggregation. What remains is consumption goods plus capital goods, so $Y \equiv C + I_g$. Output left unsold goes into inventories. The change in inventories (CII) is part of production *and* of investment, so the identity still holds, and so does $I_g \equiv S$.

Numerical template — Lecture 2 illustration

Sales 18,000, made up of inputs 3,000, consumption goods 14,000 and capital goods (tractors) 1,000. Payments: rent 1,200, interest 900, wages & salaries 8,900.

Production $Y = 18,000 - 3,000 = 15,000 = C(14,000) + I(1,000)$. Income = $8,900 + 1,200 + 900 + \text{profit } 4,000 = 15,000$, where profit is the residual.

Twist: households spend only 13,500. Then $S = 15,000 - 13,500 = 1,500$, the 500 unsold goes into CII, and $I_g = 1,000 + 500 = 1,500 = S$.

Symbolic. Sales V , inputs N , consumption goods C_g , capital goods K : $Y = V - N = C_g + K$. If households spend $C < C_g$: $\text{CII} = C_g - C$, $S = Y - C$, and $I_g = K + \text{CII} = S$.

Quick check from the slides — which are true?

(a) GDP can be measured by production, income and expenditure ✓; (b) production approach = GVA of all units ✓; (c) income approach = sum of incomes of all households ✓; (d) expenditure approach = sum of *all* expenditure on goods and services ✗, because that includes IC and only final expenditure counts; (e) total expenditure = FCE + GCF + exports net of imports ✓.

2 Economic flows and transactions

Thrust areas T4, T5, T6, T25 · Lecture 3 · PYQ: 2025–26 Q1(viii)

An **economic flow** is any change in the volume or value of a unit's economic assets. It can be caused by human acts, by changes in prices, or by natural events. Examples are production, sale and purchase, taxes, transfers, destruction by calamity, and holding gains.

Economic flows

- **Transactions:** payments and receipts for production, consumption and accumulation, and transfers.
 - **Exchange:** something is given in return.
 - **Transfer:** nothing is given in return.

• Other flows

- *Other changes in volume:* economic value appears or disappears (discoveries, destruction by calamity).
- *Changes in the level and structure of prices:* holding gains and losses.

A **transaction** is an economic flow in which institutional units interact by mutual agreement. It can also be an action within one unit that is analytically useful to treat as a transaction, usually because the unit is acting in two capacities. Transactions are either actual (observed) or notional (not observable, so imputed). They cover production and exchange of products and non-produced assets, distribution of income, transfers (current and capital), and financial assets and liabilities.

Current vs capital transactions (asked: 2025–26 Q1(viii))

Current transactions relate to production, final consumption and current transfers: production-related flows (output, IC, CFC), primary income (CE, OS/MI), production taxes less subsidies, final consumption, and current transfers such as income taxes and transfers in money or kind. They do not change the stock of assets directly and are recorded in the *current accounts*.

Capital transactions create or change the volume of a unit's assets: capital formation, acquisition and disposal of non-produced (natural) assets, capital taxes and capital transfers. They are recorded in the *accumulation accounts*, and they change the balance sheet.

Other perspectives (T25). A transaction is *between* units or *within* one unit; within-unit transactions are always non-monetary. It is *monetary* if money changes hands, and a monetary transaction always has two parties. *Non-monetary* transactions have no money flow, so they cannot be observed and are imputed, usually at market value. See §16 for the compiler's tree.

Quick check from the slides — economic flows? transactions?

Economic flow? Paddy cultivation ✓; destruction of property by a calamity ✓; father giving money to his son in the same household × (within one unit); drawing river water to irrigate land ✓; natural growth of fish stock in a river × (not an economic asset); price appreciation of real estate ✓.

Transaction: exchange or transfer? Production of crops: exchange. Income tax: transfer. Sale of smuggled goods: exchange (illegal trade still counts). Pickpocketing: transfer. Paying a lost bet: transfer.

3 The production boundary and IPP

Thrust areas T7, T8 · Lecture 4 · PYQ: —

Production (SNA) is a physical process, under the control and responsibility of an institutional unit, that uses labour and assets to turn inputs of goods and services into outputs. The output must be capable of being supplied to another unit, with or without charge. There are three kinds of product. *Goods* are physical, in demand, and carry ownership that can be transferred. *Services* change the condition of the consumer or help exchange products or financial assets (traders, banks). *Data* (2025 SNA) is information produced by observing phenomena and stored digitally; it is capitalised if used for more than a year, and then depreciated.

General production boundary: every activity that one unit could pay another to do for it, the “third-person” test. It excludes only basic human activities such as eating, drinking, sleeping and taking exercise, which nobody else can do on your behalf.

SNA (2008/2025) production boundary — inclusion and exclusion

Included	Excluded
All <i>goods</i> , whether for the market or for own final use (a farmer’s crops eaten at home).	Own-account <i>household services</i> by unpaid members: cooking, cleaning, care, repair of own durables, transporting members.
<i>Services</i> only if they are sold, supplied to other units, or produced by paid labour (paid domestic staff).	Volunteer, social and cultural activities.
Housing services of owner-occupiers (imputed rent); own-account IPP.	Things that only affect the quality of life.
Non-market output of government and NPISHs; bank services not explicitly charged (FISIM).	<i>Theft</i> (a transfer, not production).
Illegal and concealed production (narcotics trade); natural growth of <i>cultivated</i> forests.	Natural growth that nobody manages (wild fish, wild forests).

Intellectual property products (T8) are the results of research, innovation and artistic work. They have been fixed assets since the 2008 SNA. There are four categories: (1) research and development; (2) mineral exploration and evaluation; (3) computer software, data and databases; (4) entertainment, literary and artistic originals.

Quick check from the slides — inside the SNA boundary? is it production?

Inside? Maintenance and repair of household durables by the household ×; crops kept for own consumption ✓; owner-occupied housing services ✓; transporting household members ×; natural growth of cultivated forests ✓; distribution of narcotics ✓.

Production? A morning cup of tea × (nothing is created); ploughing ✓; newspaper delivery ✓ (trade); repairing a vending machine ✓ (adds value); natural growth of fish in a river × (no human effort); collecting wild honey ✓ (human effort).

4 The domestic economy: territory, units, residence

Thrust areas — (background for GNI, exports and imports) · Lecture 5 · PYQ: 2025–26 Q1(i)

- The **economy** is all institutional units *resident* in the economic territory during the period.
- The **economic territory** is the area the government administers, within which persons, goods and capital circulate freely. It includes airspace, territorial waters and continental shelf; the country's enclaves abroad (embassies, consulates, military bases, scientific stations); and free zones and offshore activity.
- An **institutional unit** can own assets, incur liabilities, carry out economic activity and transact with others in its own right. All transactions take place through institutional units. A household that also produces is treated as two units, one producing and one consuming.
- There are **five sectors**: non-financial corporations, financial corporations, general government, households (including unincorporated enterprises and the informal sector), and NPISHs. Government enterprises run like businesses (railways, airlines, power) are quasi-corporations and belong in the corporate sector. The central bank is a financial corporation.
- **Residence** is the territory where a unit has its *centre of predominant economic interest*. It does not depend on nationality or currency. An individual takes the residence of their household, and an unincorporated enterprise that of its owner. A corporation or NPI is resident where it is registered, and a foreign branch is a resident quasi-corporation of the host country. Land and buildings owned by non-residents are assigned a notional resident unit. Non-residents that transact with residents form the **rest of the world (RoW)**.
- **Exports and imports** are changes of economic ownership between a resident and a non-resident.

Quick check from the slides — export, import or neither (for India)? true or false?

A Kolkata resident buys a Swiss watch in a Kolkata shop: **N** (the shop is resident). Online purchase of a UK book: **M**. An Indian magazine delivered to an Indian student at Oxford: **N** (the student is still resident). An Egyptian's hotel bill in Mumbai: **X**.

Foreign students staying three years are residents **×**. A Citibank branch in Kolkata is resident in India **✓**. The Australian crew of a Japanese ship are residents of Japan **×**. Non-residents are not treated as owners of immovable assets **✓**. Unincorporated businesses without separate accounts belong to households **✓**. A foreign NPI's branch serving residents is a resident NPISH **✓**. The central bank is in general government **×**. A Japanese firm's branch in Thailand is resident in Thailand **✓**.

5 Economically significant prices and the three kinds of output

Thrust areas T9, T13, T14 · Lectures 5, 7 · PYQ: —

Economically significant prices are prices that significantly affect how much producers are willing to supply and how much purchasers want to buy. In practice, sales cover the majority of production costs. General government, the central bank and NPISHs mostly charge nothing or prices that are not economically significant.

Output (2025 SNA) is the goods and services an establishment produces. It excludes goods and services used in an activity where the establishment bears no risk, and those the establishment consumes itself, unless they are used for own capital formation or own final consumption.

	Market output	Non-market output	Output for own final use
What	Sold at economically significant prices	Supplied free, or at prices that are not economically significant, to other units or the community	Kept by the producer for its own final consumption or capital formation
Who	Corporations, households	General government, NPISHs, central bank (2025 SNA)	Any producer
Valued at	Quantity $\times$ market price	Sum of costs	Market-equivalent price, else sum of costs
Examples	Rented housing, transport corporations, restaurants	Police, free schools run by trusts, government hospitals	A farmer's own crops, a retailer's own shelves, owner-occupied housing, paid domestic staff, CII kept for these uses

Sum of costs (2025 SNA) = IC + remuneration of employees + depreciation (and depletion) + other taxes less subsidies on production + net return to non-financial assets + rent on non-produced non-financial assets. Products end up in one of three uses: IC, consumption of fixed capital (CFC), or **final use**, which is final consumption or capital formation.

Quick check from the slides — economically significant? which kind of output?

ESP? Entry ticket to a government museum $\times$; a pen at the local store $\checkmark$; a restaurant meal $\checkmark$; a token OPD registration fee at a free government hospital $\times$; compulsory payments by financial corporations to the central bank $\times$.

Kind? Police: non-market. Shelves a retailer builds for his shop: own final use. Owner-occupied housing: own final use. Rented housing: market. Free schools run by trusts: non-market. Transport corporations: market.

6 Basic, producers' and purchasers' prices

Thrust areas T11 · Lecture 6 · PYQ: 2025–26 Q1(iii)

A user usually pays more than the producer receives, because of taxes, trade margins and transport costs. The SNA therefore fixes three valuations:

- **Purchasers' price:** what the purchaser pays to take delivery at the time and place required, including trade and transport margins (TTM) and taxes.
- **Producers' price:** what the producer receives from the purchaser, excluding separately invoiced VAT and margins.
- **Basic price:** what the producer finally keeps, i.e. net of taxes on products and including subsidies on products.

The price chain

Purchasers' price – TTM (and taxes on consumers) = Producers' price

Producers' price – (taxes – subsidies) on products = Basic price

Output is recorded at **basic** prices; uses (IC and final use) at **purchasers'** prices. The 2008 SNA avoids the

term “market prices” for value added, because VAT makes it ambiguous. Write $GDP = GVA_{bp} + (t - s)_{\text{products}}$ rather than “GDP at market prices”.

Numerical template — Lecture 6 pen

A consumer pays 20 to a trader. The trader bought the pen at 12 from the manufacturer, who paid a product tax of 2. Purchasers' price = 20; TTM = $20 - 12 = 8$; producers' price = 12; basic price = $12 - 2 = 10$.

Symbolic. Consumer pays p , trader's cost q , product tax t , product subsidy s : TTM = $p - q$; producers' price = q ; basic price = $q - t + s$.

Numerical template — Lecture 7 Thai school

A fee of 110 Baht a month from each of 150 students, with a 10% product tax on the pre-tax fee, so $110 = 100 + 10$. GVO at producers' prices = $150 \times 110 = 16,500$; at basic prices = $150 \times 100 = 15,000$.

Symbolic. n students, fee f including tax at rate τ : producers' = nf , basic = $nf/(1 + \tau)$, tax = $nf\tau/(1 + \tau)$.

7 External transactions and trade exclusions

Thrust areas T12 · Lecture 6 · PYQ: —

Transactions with the RoW are: exports and imports; primary income (CE and investment income); current transfers, including taxes on income and wealth; capital transfers; acquisitions less disposals of valuables and non-produced assets; and financial assets and liabilities.

- **Valuation.** Goods are valued **f.o.b.** (free on board: excluding insurance and freight beyond the exporter's border) at the change of ownership. Customs record imports **c.i.f.** (cost, insurance, freight), but national accounts record imports f.o.b. too.
- **Coverage** includes non-residents' spending in the economy (tourists count as exports), services such as transport, insurance and hotels, service charges for processing goods owned by others, and **merchanting**. In merchanting, a merchant in X buys goods from Y and sells them to Z ; ownership changes, but the goods never enter X .

External trade exclusions (starred in the notes)

Rule: no change of economic ownership between a resident and a non-resident means no export or import. Excluded: (1) transactions of non-resident units with the RoW; (2) flows of samples and goods for exhibition; (3) goods sent for repair or for rent; (4) goods of returning residents; (5) goods sent for processing; (6) factor services (labour, financial resources), which are primary income, not trade; (7) goods under operational leasing or rent.

Trap — processing

The *goods* sent abroad for processing are excluded, because the owner never changes. The processing *fee* is included, as an export of services by the processor's country.

8 Factor incomes, IC, GVA and operating surplus

Thrust areas T1, T15, T17 · Lectures 5, 7 · PYQ: 2025–26 Q2(i)

A production process uses labour, inputs (IC), capital goods and natural resources. Part of the capital goods wears out (depreciation), and part of the non-renewable resources is used up (depletion). The factors of production are owned, directly or indirectly, by households:

Factor	Factor payment (compensation)
Labour	Compensation of employees, CE (“remuneration of employees”, RE, in 2025 SNA)
Land and natural resources	Rent
Capital (financial)	Interest
Entrepreneurship	Profit
A mix of labour and other factors	Mixed income (MI): earnings of unincorporated household enterprises

Rent, interest and profit together make up the **operating surplus**.

- **CE / RE**: the total remuneration, in cash or kind, an enterprise owes employees for the period’s work. It includes wages, salaries, commissions and benefits, *and* the employer’s social security and PF contributions.
- **Operating surplus (OS)** is the residual once all costs are deducted from output: $OS = GVO_{bp} - IC - CE - \text{other taxes on production (less subsidies)}$. Deduct D&D as well for net OS. OS contains the producer’s profit plus the rent and interest paid for land and funds owned by others.
- GVA_{bp} of corporations and households comes from market output plus output for own final use. That of government and NPISHs comes from non-market output plus own final use. For the whole economy, $GDP = GVA_{bp} + (t - s)_{\text{products}}$.

Numerical template — Lecture 7 bread unit → asked as 2025–26 Q2(i)

Bread worth 15,000 (without taxes), made from flour 10,000; salaries 1,000; electricity, fuel and incidentals 2,000.

$GVO_{bp} = 15,000$; $IC = 10,000 + 2,000 = 12,000$; $GVA_{bp} = 3,000$; $OS = 3,000 - 1,000 = 2,000$.

Symbolic (the exam version: garments X_1 , fibres X_2 , salary X_3 , electricity etc. X_4): $GVO_{bp} = X_1$; $IC = X_2 + X_4$; $GVA_{bp} = X_1 - X_2 - X_4$; $OS = X_1 - X_2 - X_3 - X_4$.

Trap — what goes into IC

Salaries are a *factor payment*, never IC. “Worth (without taxes)” means basic prices. **Factor service, IC or capital good?** Plant and machinery the enterprise owns: capital good. An office building it hires: IC, since the rental buys a service. Wheat in a flour mill: IC. Hired land: factor service (rent). A shareholder: factor service. Maintenance of owned machinery: IC.

9 Depreciation, depletion and gross vs net

Thrust areas T10 · Lecture 5 · PYQ: 2025–26 Q1(vi)

	Depreciation (CFC; called “depreciation” in 2025 SNA)	Depletion
Asset	Produced <i>fixed</i> assets	Non-produced <i>natural</i> resources (minerals, fuels, wild stocks)
Cause	Wear and tear and normal obsolescence from use in production	Physical removal and using up by extraction
Measures	Cost of fixed capital used up in the period	Natural resources used up in the period
Valued at	<i>Current</i> prices of the assets at the time of production, not book value	Decline in the resource’s value
In SNA	A cost of production	A cost of production (2025 SNA)

Both reduce the potential to draw benefits from the stock in future. Depreciation can be made good by new investment; production cannot re-create a depleted natural resource.

Gross – Net = D&D, for every SNA aggregate

$$\begin{aligned} \text{NVA} &= \text{GVA} - \text{D\&D}; & \text{NDP} &= \text{GDP} - \text{D\&D}; & \text{NNI} &= \text{GNI} - \text{D\&D}; \\ \text{net OS} &= \text{gross OS} - \text{D\&D}; & \text{net profit} &= \text{gross profit} - \text{D\&D}; & \text{net saving} &= \text{gross saving} - \text{D\&D}. \end{aligned}$$

Trap — company depreciation

The depreciation in a company's books is at historic cost, so it is *not* the SNA's CFC, which is valued at current prices. This is a true/false item in Lecture 12.

10 Taxes and subsidies

Thrust areas — (not on the list, but asked) · Lecture 4 · PYQ: 2025–26 Q1(vii)

Government has three economic roles: redistributing income through taxes and subsidies, non-market production (“general government”), and market production through quasi-corporations, which sit in the corporate sector.

- **Taxes** are compulsory, unrequited payments, in cash or kind, made to government units by households, enterprises and other institutional units.
- **Subsidies** are current, unrequited payments by government *to enterprises only*, for production activity. They are based on the level of production: quantities or values of goods and services produced, sold or imported.
- **Effects.** Taxes raise market prices and leave households and enterprises with fewer resources, which lowers private consumption and capital formation. Subsidies do the opposite.

Where each kind of tax enters the accounts

Taxes on products (GST, excise, import duties)	The wedge between basic and producers' prices. Added to GVA_{bp} to get GDP. Subsidies on products are the negative wedge.
Other taxes on production	Inside GVA_{bp} . Deducted before operating surplus.
Current taxes on income and wealth	<i>Current transfers</i> . They reduce disposable income, not GDP or GNI.
Capital taxes	<i>Capital transfers</i> , in the capital account.

11 Earned (primary) income and GNI

Thrust areas T1, T16, T17 · Lecture 8 · PYQ: 2025–26 Q1(i), Q2(ii)

Household earnings are of two kinds. The first comes from taking part in production. The second covers everything else: transfers, sale of property and holding gains. Only the first is **earned (primary) income**.

- **Definition.** The income generated in production is the unit's GVA (at producers' prices). It is distributed to those who supplied factor services, as factor compensation, and to government, as production taxes less subsidies. The recipients' shares are their earned (primary) income.
- **Who receives what.** Households and financial institutions receive CE, property income (interest and dividends on lent funds, rent on natural resources) and mixed income. Government receives taxes less subsidies on production and imports.
- **Property income** is (i) investment income, meaning interest (adjusted for FISIM/IFSLD), distributed income of corporations such as dividends, and income on insurance and pension entitlements; plus (ii) **rent**, the payment for using *non-produced* assets such as land or spectrum. A **rental** for *produced* assets such as buildings and machinery is the purchase of a service, and so counts as IC.

- Property income paid and received cancels within the domestic economy. That gives the **income approach**: $GDP = RE + OS + MI + (t - s)_{\text{products}} + (t - s)_{\text{imports}}$, with RE counting only what resident enterprises pay.
- **Balance of primary income** = primary income receivable – primary income payable, worked out for a unit or a sector.
- **GNI** is the sum of balances of primary income over all residents: $GNI = GDP + \text{net primary income (CE and property income) from the RoW}$. **GDP** measures production within the territory. **GNI** measures the income of residents.

Numerical template — Lecture 8 household → asked as 2025–26 Q2(ii)

One member works in a factory and another runs a grocery. (a) Salary 5,000; (b) processed food bought for resale 10,000; (c) remittance from abroad 5,000; (d) contribution to traders' association 200; (e) income tax 500; (f) sales of processed food 15,000; (g) household provisions 4,350; (h) electricity for the grocery 150; (i) hired worker 1,000; (j) sales tax 150.

In-flow = salary + GVA of the grocery = $a + (f - b - h) = 5,000 + 4,850 = 9,850$. **Out-flow** = $i + j = 1,000 + 150 = 1,150$. Balance of primary income = 8,700.

c, d and e are transfers; g is final consumption. None of them is earned income.

Quick check from the slides — earned income? true of GNI?

Earned? Beggars' income × (transfer); smugglers' income ✓ (illegal trade); gamblers' income × (transfer); gains from buying and selling shares × (holding gain); landlord's rent on land ✓; product taxes ✓; sale of land or an existing building ×; income tax revenue × (transfer).

GNI? Includes income tax received from the RoW × (that is GNDI); measures production × (that is GDP); = $GDP + \text{net primary income from the RoW}$ ✓; includes money from a relative abroad ×; measures the earned income of residents ✓; includes receipts for consultancy to the RoW ✓; unaffected by domestic production ×.

12 Transfers, disposable income, consumption and saving

Thrust areas T1, T10, T18, T19, T21 · Lecture 8 · PYQ: 2025–26 Q1(iv)

Transfers are one-way: a unit provides goods, services, an asset or money and receives nothing in return. Capital transfers are covered in §13. **Current transfers**, other than social transfers in kind, are of three kinds:

1. current taxes on income, wealth, etc.;
2. *net social contributions and social benefits*. Contributions are paid only by households, actual or imputed, and employers' contributions pass through CE first. They are received by government and by insurance and pension funds. Benefits are received by households when an event occurs: sickness, unemployment, retirement, housing, education, family;
3. *other current transfers*: net non-life premiums and claims, transfers between government units (including foreign governments), transfers to and from NPISHs, and transfers between households, including remittances.

Insurance premiums are *partly* a payment for the insurer's service; claims are pure transfers.

Quick check from the slides — current transfers?

Entirely: corporate tax; election bonds; money sent home by a non-resident; insurance claims received by a company; a worker's union contribution. Partly: life and non-life premiums. Not current: a donation to build a school (capital transfer); royalty for a patented design (payment for a service).

Disposable income

Disposable income = balance of primary income + current transfers receivable – current transfers payable, leaving out social transfers in kind. It measures purchasing power.

GNDI = $GNI + \text{net current transfers from the RoW}$, including net taxes on income and wealth = $GDP + \text{net primary income from the RoW} + \text{net current transfers from the RoW}$.

Numerical template — Lecture 8 household, continued: the likely next variant

Transfers in = $c = 5,000$; transfers out = $d + e = 700$; net = $4,300$. Disposable income = $8,700 + 4,300 = 13,000$. Extension: saving = $DI - FCE(g) = 13,000 - 4,350 = 8,650$.

Symbolic (2025–26 letters, with S' for the grocery's sales, since the paper uses S twice): $DI = S + S' - P - PE - PHW - PS + R - PT - PI$, and saving = $DI - PH$.

Consumption is using goods and services up. It takes three forms: *completely* in production (IC), *partially* in production (CFC, i.e. depreciation), or for the direct satisfaction of needs (**final consumption**). Final consumption meets *individual* needs (a household's own wants) or *collective* needs (services to the whole community). Only general government and the central bank provide collective services.

HFCE and GFCE (asked: 2025–26 Q1(iv))

Who has FCE: households, general government, the central bank and NPISHs, but *never corporations*. $FCE = PFCE + GFCE + CBCE$, where $PFCE = HFCE + FCE$ of NPISHs.

HFCE: consumption spending by *resident* households, both in the territory (on domestic output and imports) and abroad (counted as imports). It includes own-account consumption such as crops kept and owner-occupied housing. Spending by non-residents in the territory is exports, not HFCE.

GFCE: the whole non-market output of general government is treated as consumed by government itself, even though households enjoy it. Add purchases of market goods passed free to households, e.g. food for flood victims or blankets for the homeless:

$GFCE = \text{total output of GG} - \text{own-account GCF} - \text{sales of goods and services} + \text{benefits in kind}$.

Numerical template — Lecture 8 Malaysian farmer

Mixed income 5,000; rent received 1,000. Grain kept: 100 for seed, 900 to eat. Consumption goods bought 4,000; fertiliser 500. $FCE = 900 + 4,000 = 4,900$. The seed is inventory, i.e. next year's IC, and fertiliser is IC.

With current transfers of 2,000 and CFC 500: $DI = 5,000 + 1,000 + 2,000 = 8,000$; gross saving = $8,000 - 4,900 = 3,100$; net saving = $3,100 - 500 = 2,600$.

Saving: gross saving = disposable income – FCE (GNDI – FCE for the economy); net saving = gross saving – D&D.

13 Assets, capital formation and net lending

Thrust areas T8, T20, T21, T22 · Lecture 9 · PYQ: 2025–26 Q1(v)

A **financial asset** comes from a contract between two units that gives one of them the right to a payment, or a series of payments, from the other. Currency, deposits, loans, shares and bonds are examples. A **liability** is the debtor's obligation to pay the creditor. All liabilities are financial. The payment due is the financial claim, and it ends when the debtor pays.

Non-financial assets (2025 SNA) — produced vs non-produced (asked: 2025–26 Q1(v))

Produced assets come into existence as outputs of production:

- *fixed assets*, used repeatedly in production for more than a year: dwellings, other buildings and structures, machinery and equipment, weapons systems, costs of ownership transfer on non-produced assets, and IPP;
- *inventories*: materials and supplies, work in progress, finished goods, military inventories, goods for resale;
- *valuables*, held as stores of value: precious metals and stones, antiques and art, jewellery. They are neither consumed nor used in production.

Non-produced assets come into existence other than by production: contracts, leases and licences; purchased goodwill and marketing assets.

Natural resources: land, minerals, energy resources (renewable and not), water, non-cultivated biological resources (non-produced), cultivated biological resources (produced), and radio spectra (either).

Treatment differs. Produced fixed assets *depreciate*, and acquiring them is GCF. Natural resources *deplete*, and acquiring them is recorded separately, not in GCF. The 2008 SNA introduced land improvements, weapons systems and IPP as fixed assets.

Capital formation (T21, T22)

GDCF (= GCF) = GFCF + CII + acquisitions less disposals of valuables.

- **GFCF**: acquisitions less disposals of produced fixed assets, whether new or existing, and whether bought, bartered, received as capital transfers or made on own account; *plus* costs of ownership transfer on non-produced assets; *plus* major improvements that extend asset lives. Only GFCF is “investment” in the narrow sense.
- **CII** = closing stock – opening stock of inventories: unsold finished goods, traders’ stocks, unused raw materials and work in progress.

Capital transfers transfer ownership of a fixed asset, are tied to acquiring or disposing of one, or forgo a liability. Examples are grants to build a school or temple, debt waivers and capital taxes. Gross saving plus net capital transfers is usually less than GCF, and the gap is borrowed:

$$\begin{aligned} \text{Net lending (+) / borrowing (-)} &= \text{gross saving} + \text{net capital transfers} - \text{GCF} \\ &\quad - \text{net acquisition of non-produced non-financial assets.} \end{aligned}$$

Numerical template — Lecture 9 NPI

A free-education NPI receives donations of 10,000, pays salaries of 7,000, pays no tax, and has CFC of 200. It gets a capital grant of 50,000 for a classroom costing 100,000.

Saving = 10,000 – 7,000 – 200 = 2,800; funds available = 2,800 + 50,000 = 52,800; it must borrow 100,000 – 52,800 = 47,200.

Symbolic. Donation D , salaries W , CFC C , grant G , cost K : saving = $D - W - C$; borrowing = $K - (D - W - C + G)$.

Trap — CFC is not cash

The slides’ 47,200 treats CFC as reducing the funds. Strictly, gross saving is 3,000, so SNA net borrowing = 100,000 – 3,000 – 50,000 = 47,000. In the exam, follow the slides’ working; if you have time, add a line noting that CFC is non-cash.

14 The main identities

Thrust areas T3, T23 · Lectures 2, 9 · PYQ: —

Three approaches to GDP

$$\text{Production: } \text{GDP} = \text{GVO}_{bp} - \text{IC} + (t - s)_{\text{products}} + (t - s)_{\text{imports}}$$

$$\text{Income: } \text{GDP} = \text{RE} + \text{OS} + \text{MI} + (t - s)_{\text{products}} + (t - s)_{\text{imports}}$$

$$\text{Expenditure: } \text{GDP} = \text{PFCE} + \text{GFCE} + \text{CBCE} + \text{GFCF} + \text{CII} + \text{valuables} + X - M$$

The eight numbered identities (Lecture 9), which underlie the sequence of accounts:

- [1] *Commodity balance*: $\text{GVO} + M \equiv \text{IC} + \text{PFCE} + \text{GFCE} + \text{CBCE} + \text{GFCF} + \text{CII} + \text{valuables} + X$.
Everything produced or imported is used.
- [2] *Production*: $\text{GDP} \equiv \text{GVO}_{bp} - \text{IC} + (t - s)_{\text{products}} + (t - s)_{\text{imports}}$.
- [3] *Income*: $\text{GDP} \equiv (\text{RE} + \text{OS} + \text{MI})$ generated in domestic enterprises + $(t - s)_{\text{products}} + (t - s)_{\text{imports}}$.
- [4] *National income*: $\text{GNI} \equiv [3] + \text{net RE from the RoW} + \text{net property income from the RoW}$.
- [5] *Disposable income*: $\text{GNDI} \equiv \text{GNI} + \text{net current transfers} + \text{net taxes on income and wealth from the RoW}$.
- [6] *Expenditure*: $\text{GDP} \equiv \text{PFCE} + \text{GFCE} + \text{CBCE} + \text{GFCF} + \text{CII} + \text{valuables} + X - M$.
- [7] *Saving*: Gross saving $\equiv \text{GNDI} - \text{PFCE} - \text{GFCE} - \text{CBCE}$.
- [8] *Net lending*: $\equiv \text{gross saving} + \text{net capital transfers} - (\text{GFCF} + \text{CII} + \text{valuables}) - \text{net acquisition of non-produced non-financial assets}$.

Commodity-flow identity (Lecture 2), with both sides at purchasers’ prices:

$$\text{GVA}_{bp} + (t - s)_{\text{products}} + M + \text{import duties} = \text{HFCE} + \text{GCF} + X, \quad \text{i.e.} \quad Y + M = C + I + X.$$

The ladder of aggregates

GDP	
+ net primary (earned) income from RoW	= GNI
+ net current transfers (incl. taxes on income and wealth) from RoW	= GNDI
– final consumption expenditure	= Gross saving
+ net capital transfers from RoW – GCF – net acquisition of non-produced assets	= Net lending/borrowing
Subtract D&D from any of GDP, GNI, GNDI or saving to get the net version.	

Trap — D&D in the last rung

The Lecture 9 summary slide also subtracts D&D in the final step. That is only right if you start from *net* saving and use *net* capital formation (GCF – D&D). Starting from gross saving, do *not* subtract D&D. The Lecture 12 capital account confirms this: gross saving $33 + 10 = 43$, and $43 + 1 - 2 - 23 = 19$.

15 The sequence of accounts and balance sheets

Thrust areas T24 · Lecture 10 · PYQ: —

The SNA records all flows and stocks. Accounts are compiled for each of the five sectors, the RoW and the total economy. They are consistent because there is one production boundary and one asset boundary. There are three classes:

1. **Current accounts** record production, income and its use. They do not change the balance sheet directly. They are the *production account* and the *income accounts*: generation of income, allocation of primary income, secondary distribution (income transfers other than social transfers in kind), and use of disposable income.
2. **Accumulation accounts** record changes in assets and liabilities: the *capital account* (capital formation, capital transfers, non-produced assets) and the *financial account* (all transactions in financial assets and liabilities). Other changes in volume and revaluation also belong here but are not in this course.
3. **Balance sheets** record stocks at the opening and closing of the period. Assets (produced, non-produced, financial) sit on the left; liabilities (all financial) and net worth on the right. **Net worth** = total assets – total liabilities, the measure of wealth. Closing balance sheet = opening balance sheet + transactions + other flows.

Account type	Left side	Right side
Current accounts	Uses (“expenditures”)	Resources (“revenues”)
Accumulation accounts	Changes in assets	Changes in liabilities and net worth
Balance sheet	Assets	Liabilities and net worth

Every account is a T-account under double entry, so its two sides total the same. The **balancing item** is resources minus uses, shown on the uses side. It is carried forward as the first resource of the next account. The balancing items are the headline aggregates: GDP/NDP, OS/MI, GNI/NNI, disposable income, saving, and net lending/borrowing. The full worked chain is in §16.

16 Monetary transactions and the MTA

Thrust areas T25, T26 · Lectures 11–12 · PYQ: —

The compiler’s view. Two-party transactions can usually be observed. Intra-unit transactions, some non-produced items, and *composite* financial transactions (one payment that bundles several analytically different transactions) cannot, so they are imputed or derived.

Monetary transactions: money changes hands, always between two units, and they are recorded as observed. The exception is composite ones such as nominal interest and total insurance premiums, which are split into components (FISIM).

- Goods and services, produced or non-produced (g&s $\leftrightarrow$ money);
- distributive: earned income (factor services $\leftrightarrow$ money) and transfers (money $\rightarrow$ nothing);
- financial (money $\leftrightarrow$ money), simple or composite.

Non-monetary transactions: no money flows, so they cannot be observed. They are imputed at market-equivalent prices, or at the sum of costs if there is no market.

- *Two-party:* barter and payments in kind; transfers in kind;
- *intra-unit:* production for own final use (own-account capital formation, own consumption), and D&D.

Money balance identity

$$\text{money in-flow} \equiv \text{money out-flow} + \text{change in cash in hand}$$

Wallet: starts with 100; lunch 20, transport 30, client pays 150, loan from a friend 350, stationery 500.

In-flow 500, out-flow 550, change -50 , leaving 50.

In-flows come from selling products, receiving factor and property income, receiving transfers, *incurring liabilities* (taking a loan, issuing shares or bonds), and *disposing of financial assets*.

Out-flows go to buying products, paying income, pay-

ing transfers, *extinguishing liabilities* (repaying a loan, buying back shares), and *acquiring financial assets* (deposits, shares, bonds). Cash is a financial asset.

Numerical template — Lecture 11 household MTA

Sold produce 240, bought goods and services 200, received wages 20, paid wages 80, received remittances 10, paid transfers 2, borrowed 100. In-flow 370, out-flow 282, so financial assets rose by 88.

Expenditures		Revenues	
200	Bought products	Sold products	240
80	Paid income	Received income	20
2	Paid transfers	Received transfers	10
88	Change in financial assets (incl. cash)	Change in liabilities	100
370	Total	Total	370

- **Double entry.** Every monetary transaction affects two cells of a unit's MTA. A tractor (40) bought with a bank loan is "bought products 40" and "change in liabilities 40". Wages (10) paid into a bank account are "received income 10" and "change in financial assets 10". Each transaction has two parties, so the system as a whole records it four times (**quadruple entry**).
- **Aggregation** adds corresponding cells. Among residents only, products sold $\equiv$ bought, income receivable $\equiv$ payable, transfers receivable $\equiv$ payable, and change in liabilities $\equiv$ change in financial assets.
- **With the RoW**, the two sides no longer match item by item: products sold + imports $\equiv$ products bought + exports, and so on. The one identity that holds for *every* sector and for the RoW is

$$\begin{aligned} & \text{sub-total non-financial (revenue)} + \text{change in liabilities} \\ & \equiv \text{sub-total non-financial (expenditure)} + \text{change in financial assets,} \end{aligned}$$

so **non-financial balance = net lending = change in financial assets – change in liabilities**.

Expenditure		Revenue		Rev – Exp
1235	Bought products	Sold products	1259	24 (= X – M)
404	Paid income	Received income	400	–4
5	Paid transfers	Received transfers	4	–1
1644	Sub-total non-financial	Sub-total non-financial	1663	19
203	Change in financial assets	Change in liabilities	184	–19

This is the aggregated MTA of all residents (Lecture 11), and it shows the economy is a **net lender of 19**. The same identity holds sector by sector in the Lecture 12 table. Households: $405 + 4 = 401 + 8$, so they lend 4. RoW: $323 - 342 = -19$, so the RoW borrows the 19.

From the MTA to the SNA accounts (Lecture 12). The MTA records only monetary transactions and does not say what purchases were for. Extending it takes three steps: split it by sector; add the non-monetary transactions (own-use output 23, non-market output 43, so $GVO_{mp} = 1259 + 23 + 43 = 1325$); and break the broad items down. Use of products, 1301, splits into IC 870, FC 408 and CF 23. Income splits into RE, property income and $(t - s)$. Transfers split into current and capital. With $D\&D = 10$:

Account	Resources – uses	Balancing item
<i>Current accounts</i>		
Production	$GVO_{mp} 1325 - IC 870$	GDP 455; NDP 445
Generation of income	$NDP 445 - RE \text{ paid } 153 - \text{production } (t - s) \text{ and import duties } 130$	OS + MI (net) 162
Allocation of primary income	$162 + RE \text{ received } 152 + PI \text{ received } 118 + (t - s) 130 - PI \text{ paid } 121$	NNI 441
Secondary distribution	$441 + \text{current transfers received } 3 - \text{current transfers paid } 3$	NNDI 441
Use of disposable income	$441 - \text{final consumption } 408$	net saving 33
<i>Accumulation accounts</i>		
Capital	$33 + \text{capital transfers received } 1 - \text{paid } 2 - (GCF 23 - D\&D 10)$	net lending 19
Financial	$\text{change in financial assets } 203 - \text{change in liabilities } 184$	net lending 19

Check by expenditure: $408 + 23 + 24 = 455 = GDP$. Net lending is the *same* 19 as the MTA balance, because each non-monetary item enters both sides.

Rules of accounting (Lecture 12)

Valuation. Monetary exchanges are valued at the price the buyer actually paid. Non-monetary ones use market-price equivalents, or the sum of costs where there is no market. Exports and imports are f.o.b. Flows and stocks are both at current market value.

Timing. Products and financial assets are recorded when economic ownership changes. Primary income and transfers are recorded when they fall due (accrual). Output and IC are recorded when production takes place. Both parties must record the same value in the same period.

Quick check from the slides — Lectures 11–12 true/false

Cash in hand is not a financial asset ✗. Every monetary transaction makes at least two entries in a unit's MTA ✓. The MTA records stocks of assets ✗. Unequal MTA totals mean a missing entry ✓. Borrowing is a money in-flow ✓. In an open economy, products bought equal products sold ✗. The non-financial balance is net lending ✓. Money lent between residents cancels out ✓. The MTA records detailed production, consumption and accumulation ✗. All transaction identities fail for sector accounts ✗. The RoW account is presented from the non-residents' point of view ✓. Net lending is affected by non-monetary transactions ✗. Crop output for own use is recorded in the use-of-income account ✓. Import duties are in the production account ✓. All non-monetary transactions are valued at cost ✗. Primary income and transfers are on an accrual basis ✓.

17 Past papers: model answers

2025–26 Q1 — define and relate, any 5 of 8

$$[(2.5 + 2.5) \times 5 = 25]$$

Give all eight a look, then pick the five you can write fastest. The shape of each answer: define the first term, define the second, and give the relation, preferably as an equation.

(i) **GNI and GDP.** GDP is the value of production by resident producers in the economic territory during the period: $\sum GVA_{bp} + (t - s)_{\text{products}} + (t - s)_{\text{imports}}$. It can also be computed as $RE + OS + MI + (t - s)$, or as $PFCE + GFCE + CBCE + GCF + X - M$. GNI is the sum of the balances of primary income of all residents, i.e. their earned income. The relation is $GNI = GDP + (CE + \text{property income})$

received from the RoW – (CE + property income) paid to the RoW. GDP is a *territorial production* concept and GNI a *residents' income* concept. Remittances and income taxes from abroad are transfers, so they enter GNDI, not GNI. $NNI = GNI - D\&D$. [§1, §11]

(ii) **GVA and GVO.** *GVO* is the value of the goods and services an establishment produces in the period. Market output is valued at basic prices (quantity $\times$ price); non-market and own-use output at the sum of costs. It includes changes in inventories of finished goods and work in progress, and own-account capital formation. $GVA = GVO - IC$, the value the unit itself adds. It is shared out as CE + OS/MI + other net taxes on production. Summing GVA_{bp} over residents and adding net product taxes gives GDP; $NVA = GVA - D\&D$. [§1, §8]

(iii) **Basic, producers' and purchasers' prices.** Define all three (§6) and give the chain. Purchasers' price – TTM = producers' price. Producers' price – (taxes – subsidies) on products = basic price. Output is valued at basic prices and uses at purchasers' prices. Illustrate with the pen: 20, then 12, then 10.

(iv) **HFCE and GFCE.** Use the definitions in the box in §12. The relation: both are parts of FCE, with $FCE = PFCE + GFCE + CBCE$, and they enter $GDP = PFCE + GFCE + CBCE + GCF + X - M$. Households *consume* most government output, but the *expenditure* is recorded as government's. Corporations never have FCE.

(v) **Produced and non-produced assets.** Produced assets arise from production: fixed assets, inventories, valuables. Non-produced assets arise otherwise: natural resources such as land, minerals and water; contracts, leases and licences; goodwill. The treatment differs. Produced fixed assets depreciate and their acquisition is GCF. Non-produced natural resources deplete and their acquisition is a separate capital-account item. [§13]

(vi) **Depletion and depreciation.** Use the table in §9: which asset each applies to, what causes it, current-price valuation, and that both are costs of production. Then state Gross – Net = D&D.

(vii) **Taxes and subsidies.** Give both definitions from §10: compulsory and unrequited, versus current unrequited payments to enterprises only, based on the level of production. Then the relation: product $(t - s)$ is the gap between producers' and basic prices, and $GDP = GVA_{bp} + (t - s)_{\text{products}}$. Income taxes are current transfers, and taxes raise prices while subsidies lower them.

(viii) **Transactions of current and capital nature.** Use the box in §2. Current transactions involve production, primary income, final consumption and current transfers, and go in the current accounts. Capital transactions involve capital formation, non-produced assets, capital transfers and capital taxes, go in the accumulation accounts, and change the balance sheet.

2025–26 Q2 — answer one

[5]

(i) **Textile unit.** Garments X_1 (without taxes), fibres X_2 , salaries X_3 , electricity and other incidentals X_4 . [1 + 2 + 2]

$$GVO_{bp} = X_1 \quad (\text{"without taxes" means basic prices})$$

$$IC = X_2 + X_4$$

$$OS = GVA_{bp} - CE = (X_1 - X_2 - X_4) - X_3 = X_1 - X_2 - X_3 - X_4$$

State the assumptions: no other taxes on production, and OS is gross, i.e. before depreciation. Salaries are a factor payment, not IC.

(ii) **Household.** The paper uses S both for salary (a) and for sales (f). Call the sales S' and *say so in your answer*. [3 + 2]

$$\text{In-flow} = S + (S' - P - PE) \quad (\text{salary} + \text{GVA of the grocery})$$

$$\text{Out-flow} = PHW + PS \quad (\text{CE paid} + \text{product tax})$$

Excluded: R , PT and PI are current transfers, and PH is final consumption. The balance of primary income is $S + S' - P - PE - PHW - PS$. The slide numbers give 9,850, 1,150 and 8,700.